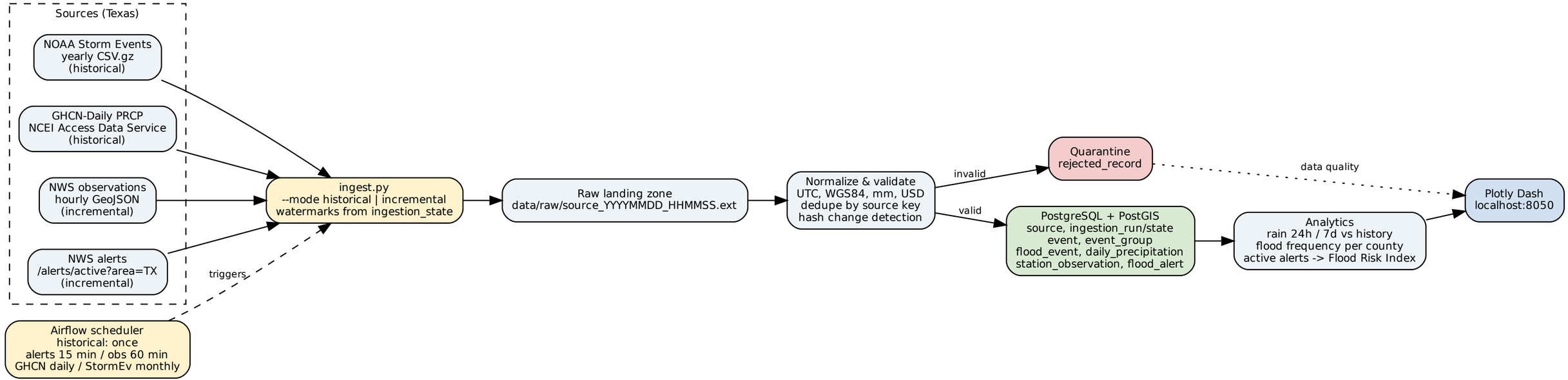
Sources (Texas)
NOAA Storm Events
yearly CSV.gz
(historical)
GHCN-Daily PRCP
NCEI Access Data Service
(historical)
NWS observations
hourly GeoJSON
(incremental)
NWS alerts
/alerts/active?area=TX
(incremental)
triggers
Airflow scheduler
historical: once
alerts 15 min / obs 60 min
GHCN daily / StormEv monthly
ingest.py
--mode historical | incremental
watermarks from ingestion_state
Raw landing zone
data/raw/source_YYYYMMDD_HHMMSS.ext
Normalize & validate
UTC, WGS84, mm, USD
dedupe by source key
hash change detection
invalid
valid
Quarantine
rejected_record
data quality
PostgreSQL + PostGIS
source, ingestion_run/state
event, event_group
flood_event, daily_precipitation
station_observation, flood_alert
Analytics
rain 24h / 7d vs history
flood frequency per county
active alerts -> Flood Risk Index
Plotly Dash
localhost:8050